

ERDENESANT, Mongolia

WMO: 442860

Lat: 47.3514N

Lon: 104.4917E

Elev: 7156

StdP: 11.27

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-18.2	-28.9	1.4	-20.8	-25.5	1.7	-17.2	23.0	1.3	21.6	3.1	4.0	180	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	86.1	57.7	82.4	57.1	79.2	56.1	62.7	77.2	61.0	74.3	59.3	72.5	8.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.6	96.1	67.3	56.5	89.2	65.9	54.7	83.4	64.9	32.3	76.7	30.8	74.3	29.4	72.2	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	21.6	19.0	DB	-26.7	91.0	4.3	3.9	-29.7	93.8	-32.2	96.1	-34.6	98.2	-37.7	101.0	(4)
(5)			WB	-27.0	65.2	4.2	2.6	-30.0	67.1	-32.5	68.6	-34.9	70.0	-37.9	71.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	-0.6	4.6	23.1	39.2	50.5	61.1	65.9	62.0	51.7	36.3	17.0	4.1		
	DBStd	25.22	9.69	11.45	11.82	10.20	10.05	7.99	6.43	6.71	9.15	9.19	11.54	10.14		
	HDD50	7082	1570	1272	834	344	122	8	0	3	87	429	989	1424		
	HDD65	11249	2035	1692	1299	774	459	165	66	137	406	888	1439	1889		
	CDD50	1510	0	0	0	20	136	341	494	376	137	6	0	0		
	CDD65	202	0	0	0	0	9	48	95	45	5	0	0	0		
	CDH74	2188	0	0	0	1	147	591	881	485	83	0	0	0		
	CDH80	596	0	0	0	0	28	167	267	121	13	0	0	0		
	WSAvg	8.0	7.7	7.5	8.5	9.9	10.5	8.5	7.0	6.7	7.3	7.7	7.5	7.4		
	PrecAvg	9.1	0.1	0.0	0.2	0.2	0.7	1.6	2.6	2.3	0.8	0.3	0.2	0.2		
(15) Precipitation	PrecMax	14.1	0.2	0.2	0.5	0.6	1.8	3.5	4.5	3.9	2.3	0.8	0.7	0.6		
	PrecMin	4.4	0.0	0.0	0.0	0.0	0.1	0.4	0.7	0.6	0.1	0.1	0.0	0.0		
	PrecStd	2.8	0.1	0.1	0.1	0.2	0.5	0.8	1.2	1.2	0.6	0.2	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	38.7	56.8	71.2	84.6	91.0	91.9	87.7	81.6	65.6	50.4	32.1		
		MCWB	21.6	28.7	38.9	45.7	54.6	58.9	61.1	59.8	53.7	44.3	36.6	25.3		
	2%	DB	22.6	32.3	50.1	66.2	78.6	84.6	86.6	83.2	75.9	60.4	44.0	26.1		
		MCWB	17.6	24.5	35.2	43.8	50.8	56.7	59.0	57.8	52.1	42.0	32.9	21.0		
	5%	DB	18.3	27.2	44.9	61.5	74.1	80.5	82.4	79.6	71.8	55.9	38.6	22.0		
		MCWB	14.1	21.2	33.0	41.8	49.3	54.9	58.4	56.8	51.0	40.0	29.6	18.0		
	10%	DB	13.6	22.3	40.4	57.0	69.1	76.8	78.8	75.5	67.5	51.5	33.5	18.1		
		MCWB	10.4	17.9	30.5	39.7	47.2	53.8	58.2	56.2	48.5	37.9	26.3	14.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	28.8	40.7	48.3	56.0	61.9	66.5	64.8	56.3	45.5	37.1	25.5		
		MCDB	28.2	38.2	55.7	68.3	80.0	82.1	78.0	81.1	74.2	63.0	49.3	32.0		
	2%	WB	17.6	24.6	36.1	44.6	52.9	59.6	63.7	61.6	53.9	42.6	33.2	21.2		
		MCDB	22.2	32.0	48.4	63.6	74.3	77.1	72.2	74.0	69.5	59.1	43.2	26.1		
	5%	WB	14.1	21.4	33.2	42.5	50.2	58.0	62.0	59.8	52.2	40.4	29.8	18.0		
		MCDB	18.1	27.0	44.5	59.9	70.7	73.9	75.8	71.4	67.9	54.7	37.8	21.8		
	10%	WB	10.4	17.9	30.6	40.3	47.9	56.2	60.4	58.3	50.4	38.2	26.7	14.8		
		MCDB	13.4	22.4	39.8	56.2	68.1	71.0	73.3	70.1	65.4	50.7	33.0	18.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	19.5	20.4	21.3	22.5	20.7	18.6	19.4	21.5	20.6	17.5	15.3		
		MCDBR	21.2	24.0	24.0	27.7	27.6	26.1	24.7	25.8	26.3	26.5	22.1	18.3		
		MCWBR	17.5	18.6	14.1	13.7	12.6	10.5	9.2	10.1	11.1	14.4	14.3	15.0		
		MCDBR	20.9	23.4	23.4	26.1	25.5	21.7	20.0	19.9	23.9	25.2	20.9	18.2		
	5% WB	MCWBR	17.3	18.4	14.2	13.5	12.8	9.7	9.1	9.5	10.9	14.1	13.9	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.210	0.229	0.264	0.298	0.319	0.331	0.321	0.304	0.269	0.248	0.226	0.212		
	taud		2.424	2.412	2.390	2.360	2.336	2.387	2.460	2.486	2.533	2.529	2.471	2.417		
	Ebn at Noon		293	307	308	304	299	295	297	298	303	296	283	278		
	Edh at Noon		20	26	31	36	38	37	34	31	27	23	19	18		
(44) All-Sky Solar Radiation	RadAvg		607	967	1436	1829	2137	2144	2036	1790	1515	1039	644	485		
	RadStd		48	62	71	73	120	128	102	108	63	54	49	33		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	-5.04	N/A	-1.19	-1.58	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page